

Exponential vs. Linear Growth

DSAT Math Practice | 25 Questions | Difficulty-Graded | ~15-20% Grid-Ins

Directions: Answer each question by selecting the best answer from the choices given, or by entering your answer in the grid-in box for student-produced response questions. You may use a calculator on all questions. Reference the provided formulas as needed.

1

Easy

A savings account starts with \$500 and earns \$40 in interest each year. Which function models the amount A , in dollars, after t years?

A

$$A(t) = 500(1.08)^t$$

B

$$A(t) = 500 + 40t$$

C

$$A(t) = 40(500)^t$$

D

$$A(t) = 500(40)^t$$

2

Easy

A population of bacteria doubles every 3 hours. If the initial population is 200, which function models the population P after t hours?

A

$$P(t) = 200 + 2t$$

B

$$P(t) = 200(2)^{t/3}$$

C

$$P(t) = 200(3)^{t/2}$$

D

$$P(t) = 200(2)^{3t}$$

3

Easy

The table shows the value of an investment over 4 years. Year 0: \$1,000; Year 1: \$1,060; Year 2: \$1,123.60; Year 3: \$1,191.02. Which type of growth does this represent?

A

Linear growth, because the differences are constant.

B

Linear growth, because the ratios are constant.

C

Exponential growth, because the differences are constant.

D

Exponential growth, because the ratios are constant.

4

Easy

Which of the following is an exponential function?

A

$$f(x) = 3x + 7$$

B

$$f(x) = 7(3)^x$$

C

$$f(x) = 3x^2 + 7$$

D

$$f(x) = 7x^3$$

5

Easy

A car depreciates in value by 15% each year. If the car is currently worth \$24,000, which expression gives its value after 5 years?

A

$$24000(0.15)^5$$

B

$$24000(0.85)^5$$

C

$$24000 - 0.15(5)$$

D

$$24000(1.15)^5$$

6

Easy

A plant grows 3 centimeters per week. If the plant is currently 12 centimeters tall, what is its height, in centimeters, after 8 weeks?

[Student-Produced Response]

7

Medium

The function $f(x) = 800(0.92)^x$ models the value of a piece of equipment after x years. Which statement best describes this function?

A

The equipment loses 92% of its value each year.

B

The equipment loses 8% of its value each year.

C

The equipment gains 8% of its value each year.

D

The equipment loses \$0.92 each year.

8

Medium

A town had a population of 12,000 in 2015. The population has been increasing by 800 people per year. A nearby city had a population of 8,000 in 2015 and has been growing by 12% per year. In which year will the city's population first exceed the town's population?

A

2020

B

2022

C

2024

D

2026

9

Medium

The number of subscribers to a newsletter is modeled by $S(t) = 250(1.06)^t$, where t is the number of months since January. What does the 250 represent in this context?

A

The monthly growth rate

B

The number of subscribers in January

C

The number of subscribers after 6 months

D

The total number of subscribers gained each month

10

Medium

A lab culture starts with 500 cells and triples every 4 hours. Another culture starts with 2,000 cells and grows by 200 cells per hour. After approximately how many hours will the first culture exceed the second?

A

8

B

10

C

12

D

16

11

Medium

A radioactive substance decays by 20% every 6 years. If the initial mass is 100 grams, what is the mass, in grams, after 18 years?

[Student-Produced Response]

12

Medium

Which of the following tables represents an exponential relationship between x and y ?

A

x : 1,2,3,4 and y : 5,10,15,20

B

x : 1,2,3,4 and y : 3,9,27,81

C

x : 1,2,3,4 and y : 2,5,10,17

D

x : 1,2,3,4 and y : 4,8,14,22

13

Medium

The value of a collectible card increases by 5% each year. The value of another card increases by \$4 each year. Both cards are currently worth \$60. After how many years will the first card be worth more than the second?

A

The first card is always worth more.

B

The first card is never worth more.

C

After approximately 13 years.

D

After approximately 25 years.

14

Medium

A biologist observes that a population of insects can be modeled by $f(t) = 50(1.40)^t$ and a population of predators can be modeled by $g(t) = 200 + 60t$, where t is in months. What is the approximate value of t when $f(t)$ first exceeds $g(t)$?

A

$t = 4$

B

$t = 6$

C

$t = 8$

D

$t = 10$

15

Medium

The function $f(x) = a(b)^x$ passes through the points (0, 5) and (2, 45). What is the value of b ?

- A 3
- B 5
- C 9
- D 40

16

Hard

The half-life of a certain medication in the bloodstream is 4 hours. A patient takes a 600 mg dose. Which function gives the amount, in mg, remaining after t hours?

- A $A(t) = 600(0.5)^{t/4}$
- B $A(t) = 600(0.5)^{4t}$
- C $A(t) = 600 - 150t$
- D $A(t) = 600(0.5)^{4/t}$

17

Hard

The function $f(t) = 2000(0.97)^t$ models the number of active users of a website t days after a change in policy. After how many complete days will the number of active users first drop below 1,500?

[Student-Produced Response]

18

Hard

A company offers two salary plans. Plan A: \$50,000 base with a \$2,000 raise each year. Plan B: \$45,000 base with a 5% raise each year. In which year does Plan B first offer a higher annual salary than Plan A?

- A Year 6
- B Year 8
- C Year 10
- D Year 12

19

Hard

The function $g(x) = 3(2)^x - 5$ is graphed in the xy -plane. Which of the following statements is true?

A

The graph has a horizontal asymptote at $y = 0$.

B

The graph has a horizontal asymptote at $y = -5$.

C

The graph has a vertical asymptote at $x = -5$.

D

The graph has no asymptotes.

20

Hard

The population of City A is 50,000 and grows by 3% per year. The population of City B is 80,000 and grows by 1,200 people per year. Which statement is true about the long-term populations?

A

City A will always have fewer people than City B.

B

City A's population will eventually exceed City B's population.

C

Both populations will be equal at some point and then stay equal.

D

City B's population will eventually decrease.

21

Hard

A scientist models the decay of a sample as $N(t) = N_0(0.8)^{t/5}$, where t is in years. What is the annual decay rate?

A

4%

B

4.4%

C

16%

D

20%

22

Hard

The function $f(x) = a(b)^x$ satisfies $f(1) = 18$ and $f(4) = 486$. What is the value of a ?

[Student-Produced Response]

23

Hard

Which of the following best describes the key difference between the functions $f(x) = 100 + 25x$ and $g(x) = 100(1.25)^x$ for large values of x ?

A

$f(x)$ grows faster because it starts with a larger rate of change.

B

$g(x)$ eventually grows much faster because exponential growth outpaces linear growth.

C

Both functions grow at the same rate for large x .

D

$f(x)$ approaches a horizontal asymptote while $g(x)$ does not.

24

Hard

A social media post gets 50 shares in the first hour. Each subsequent hour, the number of new shares is 70% of the previous hour's shares. Which expression gives the total number of shares after 6 hours?

A

$50(1 - 0.7^6) / (1 - 0.7)$

B

$50(0.7)^6$

C

$50 + 50(0.7)(5)$

D

$50(6)(0.7)$

25

Hard

The graph shows two functions: $f(x) = 2^x$ and $g(x) = 4x$. For what positive value of x does $f(x) = g(x)$?

A

$x = 1$

B

$x = 2$

C

$x = 4$

D

$x = 8$
